Automated Customer Support Ticket Classification Using Machine Learning, Ensemble Methods, Neural Networks, Embeddings, and Transformer Models

Dataset Link

Dataset Name: Customer Support Ticket Dataset

Source: Kaggle

Access Link: https://www.kaggle.com/datasets/suraj520/customer-support-ticket-dataset

Problem Explanation

In today's digital environment, businesses such as e-commerce companies, banks, telecommunications providers, and online service platforms receive a large number of customer support requests every day. These requests are typically submitted through emails, live chat systems, websites, or mobile applications. Each request describes a specific issue faced by the customer, such as billing errors, delivery delays, refund requests, account access problems, or technical difficulties.

Customer support teams must review each incoming message and assign it to the appropriate department for resolution. This process is commonly performed manually, requiring human agents to read and interpret the content of every ticket. As the volume of customer requests increases, manual processing becomes inefficient and difficult to manage.

Core Challenges

- High Volume of Customer Requests: Large organizations may receive thousands of support tickets daily, making manual handling time-consuming.
- Slow Response Time: Delays in categorizing tickets result in slower customer service.
- Human Error in Classification: Incorrect routing of tickets leads to inefficiencies.
- Increased Operational Costs: More staff are required to manage large ticket volumes.
- Poor Customer Experience: Delayed responses reduce customer satisfaction and trust.

Objective of the Project

The objective of this project is to develop an automated text classification system that can analyze customer support messages and assign them to the correct category using machine learning and deep learning techniques. The system will process messages automatically, identify patterns in text data, predict the appropriate ticket category, and improve response time and operational efficiency.